**2. Mid Atlantic Region**

The Mid Atlantic Region coasts has comparatively fewer active volcanoes but many dormant or extinct volcanoes, example. St. Helena, Cape Verde Islands and the Canary Islands. But the volcanoes of Iceland and the Azores are active.

**3. The Great Rift valley of Africa**

In Africa some volcanoes are found along the East African Rift Valley. Kilimanjaro and Mt. Kenya are extinct volcanoes. The only active volcano in West Africa is Mt. Cameroon.

**4. Mediterranean Region**

Volcanoes of the Mediterranean region are mainly associated with the Alpine folds. Example, Mt. Vesuvius and Mt. Stromboli (known as the Light House of the Mediterranean Sea).

**5. Other Regions**

Elsewhere in the interiors of continents of Asia, North America and Europe active volcanoes are rare. There are no volcanoes in Australia.

**Volcanoes in India**

There are no volcanoes in the Himalayan region of India. However, Barren Island, lying 135 km north-east of Port Blair became active in 1991 and 1995.

However, the other volcanic island in Indian Territory is Narcondam (Andaman and Nicobar Islands) It is probably extinct. Its crater wall has been completely destroyed.

## 3.11 Rocks

Rock is the solid mineral material forming the surface of the earth. Petrology is the science of rocks. The age of the rock is determined based on Carbon-14 dating.

### 3.11.1 Rock Types

Based on their origin, the rocks are classified as follows:

There are three types of rocks

1. Igneous Rocks
2. Sedimentary Rocks
3. Metamorphic Rocks

**I. Igneous Rocks**

Igneous rocks are formed out of magma and lava and they are known as primary rocks. If the magma cools slowly at great depths, mineral grains increase in their size. Sudden cooling (at the surface) results in small and smooth grains. The igneous rocks are the oldest of all the rocks. Granite, pegmatite, basalt, etc are some of the examples of igneous rocks. There are two types of igneous rocks: intrusive rocks (Granite) and extrusive rocks (Basalt-Deccan Traps).

Granite is less dense and is lighter in colour than basalt rocks.

### 3.11.2 Intrusive Igneous rocks

Intrusive Igneous rocks are formed when magma rises and cools within the crust. The intrusive activity of volcanoes gives rise to various forms. We see them one by one as follow.

**1. Batholiths**

Batholiths are large rock masses formed due to cooling and solidification of hot magma inside the earth. It is granitic in origin.

**2. Laccoliths**

Laccoliths are large dome-shaped intrusive rock connected by a pipe-like conduit from below. These are basically intrusive counterparts of an exposed domelike batholiths. The Karnataka plateau is spotted with dome hills of granite rocks. Most of these, now exfoliated, are examples of laccoliths.



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